

Student Performance Prediction System

A Machine Learning-based web application that predicts a student's final examination grade (G3) using academic, demographic, and personal information. The project utilizes multiple regression algorithms, compares their performance, and deploys the best-performing model through a user-friendly Streamlit interface.

Project Overview

Student performance is influenced by various factors such as study habits, parental education, previous academic records, attendance, and family support. Manually analyzing these factors for every student is difficult and time-consuming.

This project uses Machine Learning to predict a student's final grade based on these factors, allowing educators and students to identify academic performance early and take appropriate actions.

Objectives

- Predict the final examination grade (G3) of students.
 - Compare different Machine Learning regression models.
 - Select the most accurate model.
 - Build an interactive web application using Streamlit.
 - Provide personalized suggestions based on the predicted performance.
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Features

- Interactive and easy-to-use Streamlit interface.
 - Predicts the final grade on a scale of **0-20**.
 - Uses **Random Forest Regressor** for prediction.
 - Accepts multiple student-related inputs.
 - Provides personalized performance suggestions.
 - Fast and accurate prediction.
 - Simple deployment and easy maintenance.
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Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Joblib

- Matplotlib
 - Streamlit
 - Jupyter Notebook
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Dataset

Dataset Name: Student Performance Dataset

The dataset contains **395 student records** with **33 attributes** describing academic, demographic, family, and social factors.

Some important features include:

- School
 - Gender
 - Age
 - Address
 - Family Size
 - Parents' Education
 - Study Time
 - Previous Failures
 - School Support
 - Family Support
 - Paid Classes
 - Internet Access
 - Absences
 - Health
 - Going Out Frequency
 - First Period Grade (G1)
 - Second Period Grade (G2)
 - Final Grade (G3)
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Machine Learning Models Used

The following regression algorithms were trained and evaluated:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

After comparing model performance using evaluation metrics, **Random Forest Regressor** was selected as the final model.



Model Evaluation Metrics

The models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

These metrics helped determine the best-performing model for predicting student performance.



Streamlit Web Application

The Streamlit application allows users to:

- Enter student information.
 - Predict the expected final examination grade.
 - View the predicted score instantly.
 - Receive performance-based suggestions.
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Project Structure

```
StudentPerformancePrediction/  
├── dataset/  
│   └── student_data.csv  
├── notebook/  
│   └── StudentPerformancePrediction.ipynb  
├── graphs/  
│   ├── 01_G3_Distribution.png  
│   ├── 02_Correlation_Heatmap.png  
│   └── ...  
├── model/  
│   └── student_performance_model.pkl  
├── screenshots/  
│   ├── Home.png  
│   ├── Input.png  
│   └── Prediction.png  
├── app.py  
├── requirements.txt  
└── README.md
```

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├─ Project_Report.pdf
└─ Presentation.pptx
```

⚙️ Installation

Clone the repository

```
git clone <repository-url>
```

Move into the project directory

```
cd StudentPerformancePrediction
```

Install dependencies

```
pip install -r requirements.txt
```

Run the application

```
streamlit run app.py
```

📷 Application Screenshots

Add the following screenshots:

- Home Page
 - Sidebar
 - User Input Form
 - Prediction Result
 - Graphs
 - Model Output
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🚀 Future Scope

- Predict student performance using Deep Learning models.
- Integrate real-time student databases.
- Deploy the application on cloud platforms.
- Improve prediction accuracy using advanced feature engineering.
- Provide personalized learning recommendations.

- Develop a mobile application version.
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References

1. Student Performance Dataset.
 2. Scikit-learn Documentation.
 3. Streamlit Documentation.
 4. Pandas Documentation.
 5. NumPy Documentation.
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Developed By

Princy Tripathi

B.Tech – Computer Science & Engineering

United College of Engineering & Research